PREVUMO · speaker script

GNG4120 GA3 · Group 10 · Friday Oct 23, 2026 · 8:00 plus Q&A. About 130 words per minute.

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1 · Title

`SPEAKER A · 0:00 to 0:20 (20 s, about 45 words)`

Speakers: A = [FILL IN], B = [FILL IN], C = [FILL IN], D = [FILL IN], E = [FILL IN].

Picture the welder in your school's makerspace. Expensive. Shared. Booked every single day.

Everyone knows who reserved it. Nobody knows what is happening inside it.

That gap is why we built PREVUMO, the preventive usage monitor. Let me show you what that gap costs.

`CUE: click on "what that gap costs".`

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2 · Hook

`SPEAKER A · 0:20 to 0:45 (25 s, about 50 words)`

A three thousand dollar welder dies on a Tuesday.

[pause]

Nobody saw it coming.

[pause]

The owner can open the booking software and see exactly who booked it that morning. But not what happened to it, and not whether it had been showing warning signs for weeks.

`CUE: click on "warning signs for weeks".`

3 · The pain

SPEAKER A · 0:45 to 1:45 (60 s, about 130 words)

More and more makerspaces, university labs and shared kitchens now rent out expensive electric equipment. We talked to the people who run them, and three pains kept coming up.

First, repairs. A makerspace director told us a breakdown costs "hundreds if not thousands of dollars." Industry data puts the average emergency repair call at seven hundred and fifty four dollars.

Second, lost customers. When machines keep breaking, members cancel, and each member is worth six hundred to six thousand dollars to that space.

Third, the blind spot. Booking tools only work, in his words, "when people actually use the software." And no sensor is watching the machine itself. So when it fails, there is no history to learn from.

CUE: click on "no history to learn from".

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4 · Customer and value

SPEAKER A, then hand off to B · 1:45 to 2:15 (30 s, about 65 words)

So our customer is the owner or manager of a shared space with costly electric equipment: makerspaces, labs and shared kitchens. Often one owner runs several locations.

Our promise to them fits in one sentence: catch failures before they happen, know who used each machine and how much, and shut it down safely from anywhere.

TRANSITION (A): B, so how do we actually see inside the machine?

B: Let me show you.

CUE: B clicks on "Let me show you".

5 · How it works

`SPEAKER B · 2:15 to 3:00 (45 s, about 98 words)`

It starts with a small box on each machine's power line: a keypad, a current sensor and a relay, run by an ESP32 with Wi-Fi.

A member types their PIN, and only then does the machine power on. While they work, the box measures the current every second and sends it to the PREVUMO cloud.

Over its first week, the app learns that machine's normal curve. When the curve breaks pattern, the owner gets an alert and chooses: shut it down, call a technician, or mark it OK.

And two safety rules: nothing restarts on its own after a shutdown, and fridges are never cut.

`CUE: click on "fridges are never cut".`

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6 · Prototype demo

`SPEAKER B · 3:00 to 4:15 (75 s). The video is 67 s with no sound. Click the video to start it, then narrate on the cues in brackets (video time).`

Before you click: "Here is the owner app. Everything you'll see works today."

[CLICK THE VIDEO]

`[0:00]` The owner logs in. uOttawa Engineering has three locations.

`[0:06]` Three machines are running: one started with a PIN, one unassigned, one booked to a capstone team. Below, the whole day on one timeline.

`[0:11]` Now Welder 2 starts drawing more than twice its normal current.

`[0:15]` Within seconds, this banner.

`[0:18]` Review shows the live curve against this welder's normal range, in green. Orange dots are the irregular points. Third time this month, 74 days since its last service.

`[0:25]` The owner calls a technician. The best match is preselected, and the machine's history goes with the request.

`[0:34]` Meanwhile, a 3D printer has been idle for 45 minutes, still drawing power. One tap shuts it down, for good.

`[0:46]` The owner gives Beta Inc. the laser cutter for this afternoon.

`[0:56]` And at the end of the day, the report builds itself, per machine and per member, ready to export.

`[1:05]` That's the loop: warning, decision, record. Scan the QR code to try it yourself.

`TRANSITION (B): C, who else is going to want this?`

C: Let's talk about the market.

`CUE: C clicks on "the market".`

7 · Market

`SPEAKER C · 4:15 to 5:00 (45 s, about 98 words)`

Let's size our first markets.

In Ontario alone, there are thirty one shared-use commercial kitchens, each one full of ovens, mixers and fridges that run all day. And kitchens matter, because restaurants use five to seven times more energy per square foot than other commercial buildings. Idle equipment there is expensive.

The method itself is proven. The U.S. Department of Energy reports that predictive maintenance saves about thirty to forty percent compared with fixing machines after they break.

Our path: start with one makerspace, grow into university networks with several labs, then move into shared kitchens.

[FILL IN: add a sourced makerspace count here if the team finds one; otherwise leave it out.]

`CUE: click on "shared kitchens".`

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8 · Competition

`SPEAKER C, then hand off to D · 5:00 to 5:45 (45 s, about 98 words)`

So who's already out there? Every competitor knows one piece of the story.

Skedda, from about two hundred and fifty US dollars a month, knows who booked.

Fabman knows who switched the machine on.

Emporia, around two hundred dollars, knows how much energy.

MachineMetrics knows machine health, but it's built for factories, with hardware from about fourteen hundred dollars a kit.

PREVUMO puts who used it, how much, machine health and remote shutdown together, built for shared spaces.

And Skedda can be a partner: a Skedda booking becomes a PIN on our box.

`TRANSITION (C): D, so how does this make money?`

D: Here's our model.

`CUE: D clicks on "our model".`

9 · Business model

`SPEAKER D · 5:45 to 6:45 (60 s, about 125 words)`

Our business model changed a lot.

On the left is our first canvas, Plan A: a usage tracker that bills members for the electricity they use. On the right is today's canvas, Plan B: safety and predictive maintenance. The orange blocks are the ones that changed.

Revenue is a box at about three hundred dollars per machine, one time, plus a subscription of about twenty five dollars per machine per month. We took the per machine idea from Fabman's pricing. A possible extra is a referral fee on the technician jobs we send out.

To be clear, these prices are assumptions we still have to validate.

Our key partners are electricians and technicians, component suppliers, and booking platforms like Skedda.

[FILL IN: confirm with the team which BMC blocks changed between Plan A and Plan B; the slide highlights value proposition, key partners and revenue streams.]

`TRANSITION (D): E, what have we learned so far?`

E: More than we expected.

`CUE: E clicks on "more than we expected".`

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10 · Progress and validation

`SPEAKER E · 6:45 to 7:40 (55 s, about 115 words)`

Here's where we are. We reverse engineered three competitors, built two business model canvases, sent forty nine outreach emails to kitchens and makerspaces in Canada, the US and Europe, and ran [FILL IN: N] interviews.

The executive director of an Ontario makerspace told us: "Managing equipment is one of the biggest tasks we have." He also said they are "currently working on a system to track electricity usage on the welders."

So the problem is real. But one of our assumptions failed. Owners don't bill electricity per user, because it's built into membership fees. So we moved our headline from billing to predictive maintenance. Billing stays as a bonus.

[FILL IN: only name the interviewee if he agrees; otherwise keep "Executive Director, Ontario makerspace".]

`CUE: click on "Billing stays as a bonus".`

11 · Closing

`SPEAKER E · 7:40 to 8:00 (20 s, about 40 words)`

Skedda tells you who booked it.

[pause]

PREVUMO tells you who used it, and when it's about to fail.

[pause]

Thank you. We'd love your questions, and you can try the app right now with this QR code.

Appendix · Hardware

`APPENDIX, for Q&A only. Remote shutdown hardware.`

The box's own relay is certified for up to 15 A at 120 V, which covers 3D printers, laser cutters and most bench tools. For 240 V or 3-phase machines like large welders and ovens, the box drives a contactor, installed by a licensed electrician. Before we sell, the box needs CSA certification. Safety rules in software: nothing restarts on its own after a shutdown, and fridges and freezers are never cut.

Appendix · Learning, privacy, limits

`APPENDIX, for Q&A only. Learning, privacy, limits.`

The AI needs about 7 days of normal use to learn a new machine's current pattern, so alerts switch on after that. Data is private to the owner's account. And electricity data can't see gas equipment, so gas ranges and burners are outside what PREVUMO can watch.

Appendix · Validation board

`APPENDIX, for Q&A only. Full validation board.`

[FILL IN: success criteria for each row, the interview count, and the result for the second row once the team agrees.]

[FILL IN: confirm the problem wording in each row against the team's board.]

Only the first row's result is stated in our research: owners don't bill electricity per user, so that assumption is invalidated.

Appendix · Q&A prep

APPENDIX, Q&A prep. Short answers, then the fuller version.

1. "Why wouldn't owners just use a smart plug?"

A smart plug knows energy and can switch off. It doesn't know who used the machine, because there's no PIN. It doesn't learn the machine's normal pattern, so it can't warn you before a failure. And most plugs are for small 120 V loads, not a welder or an oven.

2. "What about heavy 240 V equipment?"

Our relay is certified up to 15 A at 120 V. For 240 V or 3-phase machines, the box drives a contactor installed by a licensed electrician, and we'd need CSA certification before selling.

3. "Who's liable if a remote shutdown ruins a job?"

The owner makes the call, or sets the rule. Every shutdown is logged with who decided and why, nothing restarts on its own, and fridges are never cut. Before we sell, our terms would set out responsibility, reviewed by a lawyer. [FILL IN if the team has a better answer.]

4. "How does the AI know what normal is?"

It watches each machine for about 7 days of normal use and learns its own current pattern, including duty cycles like an oven's heater. Then it flags sustained changes, a spike, a sag or erratic swings, not one noisy reading. We plan to use the owner's "mark as OK" decisions as feedback.

5. "Why would a not-for-profit makerspace pay $25 per machine?"

That's about $300 per machine per year. One avoided emergency repair call averages $754, and one member kept is worth $600 to $6,000. They could also put only their two or three most expensive machines on it. The price is still an assumption we're testing.

6. "How is this different from Fabman?"

Fabman controls access and knows who switched a machine on. PREVUMO also learns each machine's normal current, flags irregularities before a breakdown, detects idle machines and connects the owner to a technician with the machine's history.

7. "What did the interviews change?"

We started with billing members for the electricity they use. Owners told us they don't bill electricity per user, it's built into membership fees. So we moved the headline to predictive maintenance and kept billing as a bonus.